

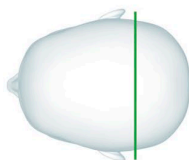
GREY AND WHITE MATTER

Most of the cerebrum has two main layers. The outer layer, often known as “grey matter”, is the cerebral cortex. It follows the folds and bulges of the cerebrum to cover its entire surface. Its average thickness is 3–5mm ($\frac{1}{10}$ – $\frac{2}{10}$ in), and, spread out flat, it would cover about the same area as a

pillowcase. Deeper within the cerebrum are small islands of grey matter. These, and the cerebral cortex, are composed chiefly of the cell bodies and projections (dendrites) of neurons. The paler “white matter” forms the bulk of the cerebrum’s interior. It is composed mainly of nerve fibres.

CORONAL SECTION

A vertical “slice” through the middle of the brain reveals the paired structures, outer grey layer, and inner white matter. The corpus callosum contains more than 100 million nerve fibres, and is the main “bridge” between the two hemispheres.



Corpus callosum

Largest of several bundles of nerve fibres, called commissures, which connect specific areas of the two halves, or cerebral hemispheres, of the upper brain

Grey matter
Outermost layer of cerebral cortex; contains an estimated 50 billion neurons and perhaps 10 times as many supporting cells

White matter interior

Here, axons, or fibres, of neurons run up from lower areas and project down from neuron cell bodies of cortex

Basal ganglia

“Islands” of grey matter deep in cerebrum

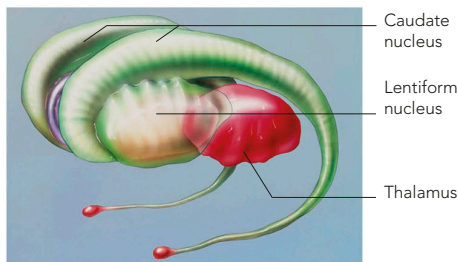
Motor nerve tracts

Large bundles of nerve fibres carry instructions for movements down to the spinal cord; they cross over in lower brainstem

Brainstem

BASAL GANGLIA

These structures include the lentiform nucleus (putamen and globus pallidus), caudate nucleus, subthalamic nucleus, and substantia nigra (the latter two not seen in this view). They are a complex interface between sensory inputs and motor skills, especially for semi-automatic movements, such as walking.



Caudate nucleus

Lentiform nucleus

Thalamus